

SuperBundle - Bundle Adjustment with Superquadric Primitives

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Abstract

1. Introduction

Reconstructing 3D scenes from multiple camera images is a classical problem in Computer Vision. There are many feed-forward models for predicting camera parameters and point clouds to reconstruct full 3D scenes [5, 12, 14]. However, especially in sparse view settings, camera pose estimates can drift or be imprecise. This comes from the limited overlap of the views, ambiguous geometry, and scale uncertainty. To reduce the prediction error one can use optimization based methods like bundle adjustment [11] to refine the reconstructed scenes.

In this project, we improved the camera poses in very sparse view settings with a modified version of the classic bundle adjustment algorithm. We used a superquadric [8] representation of our 3D scene as a structural prior, improving the camera pose estimates compared to the classic bundle adjustment. For this, we used the superquadric representation of the Aria Synthetic Environments Dataset[1] generated with Superdec [3] from the ground truth point clouds.

XXX Add stuff with connection to results and discussion

2. Related Works

Feed Forward Reconstruction Traditional multi-view reconstruction pipelines [4, 7, 9] rely on sequential stages of feature extraction, matching, and geometric optimization, making them computationally expensive and sensitive to initialization. Feed-forward reconstruction models address this by directly regressing 3D structure and camera parameters from images in a single forward pass. Feed-forward reconstruction models address this by directly re-

gressing 3D structure and camera parameters from images in a single forward pass. DUS3R [13] and MAS3R [6] established this paradigm by predicting dense point maps and relative camera poses from image pairs without requiring known intrinsics, relying on post-hoc global alignment to extend beyond pairs. Methods like VGGT [12] or MapAnything [5] advanced this further by processing up to hundreds of images simultaneously in a single forward pass, jointly predicting camera parameters, depth maps, point maps, and 3D point tracks.

Bundle Adjustment Bundle adjustment (BA) is the gold standard for refining camera poses and 3D structure in multi-view reconstruction. Formally introduced by Triggs et al. [11], it minimizes the total reprojection error over all cameras and scene points simultaneously via the Levenberg-Marquardt algorithm, exploiting the sparse block structure of the Jacobian to keep the problem tractable. In classical SfM pipelines such as COLMAP [9], BA is applied both locally after each image registration and globally as a final refinement, preventing drift and enforcing geometric consistency across the full reconstruction. While highly effective given sufficient image overlap and reliable correspondences, classical BA is sensitive to initialization and provides no mechanism to recover from fundamentally incorrect pose estimates. This limitation becomes particularly acute in the sparse-view setting.

Superquadrics Superquadrics are a family of parametric geometric primitives, first introduced by Barr [2] and later popularized for 3D shape recovery by Solina and Bajcsy [10], capable of representing a wide range of shapes — from boxes to cylinders to ellipsoids — with a small set of parameters encoding position, orientation, scale, and two shape exponents. Their compactness and analytical differentiability make them attractive as structural scene representations. SuperDec [3] builds on this foundation by pre-

senting an approach for creating compact 3D scene representations through decomposition into superquadric primitives, designing a novel architecture that efficiently decomposes point clouds of arbitrary objects into a compact set of superquadrics. Rather than targeting photorealistic reconstruction, SuperDec solves the decomposition locally on individual objects and leverages instance segmentation to scale to full 3D scenes. In our work, we use SuperDec to obtain a compact geometric abstraction of the scene from the ground truth point cloud, which we then use as a structural prior to constrain and guide bundle adjustment optimization.

3. Method

4. Results

5. Discussion

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